REVISION 2021 - PROGRAM CORE

Industrial Engineering

A complete diploma engineering handbook for production planning, plant layout, work study, statistical quality control, acceptance sampling, estimating, costing and depreciation.

COURSE CODE
4024

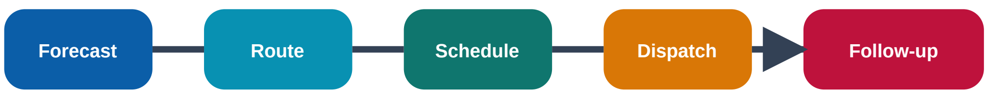
SEMESTER
4

CREDITS
No Credit

PERIODS
45

Industrial Engineering □□□□□□□□□□□□ factory flow, time, cost, quality, maintenance, inspection, productivity □□□□□□ □□□□□□□ □□□□□.

Engineering factory improvement flow



PPC converts demand into controlled shop-floor action

Course Overview

What this subject is

Industrial Engineering improves productivity, quality, cost, delivery and utilisation of men, machines, materials, methods and money. It is a factory thinking subject: it teaches how to plan work, arrange machines, measure time, control quality, estimate cost and make acceptance decisions.

Why it matters

A production line fails not only because of machine breakdown. It also fails because of poor routing, wrong layout, missing tools, excess movement, loose quality control, wrong standard time and inaccurate costing. Industrial engineering identifies these losses and removes them systematically.

How to study

1. Learn every term with a shop-floor example.
2. Draw all charts and curves neatly.
3. Practise numerical problems: standard time, mean, range, cost and depreciation.
4. For every answer add one application in production, maintenance, panel assembly, tool room or inspection.

Official information

Item	Information
Program	Diploma in Mechanical Engineering / Tool and Die Engineering / Manufacturing Technology
Course Code	4024
Course Title	Industrial Engineering
Semester	4
Credits	No Credit
Category	Program Core
Periods	3 per week, 45 per semester
Practicals	Not specified in supplied syllabus/source.

Course Outcomes

CO	Outcome	Hours	Level	Engineering meaning
CO1	Describe PPC, plant layout and plant maintenance.	11	Understanding	Plan, route, schedule, dispatch and maintain a production system.

CO2	Apply method study and work measurement techniques.	10	Applying	Improve work method and fix standard time.
CO3	Interpret control charts used in quality control.	12	Applying	Use statistics and charts to judge process stability.
CO4	Explain acceptance sampling, selling price and depreciation.	10	Applying	Connect inspection risk, cost structure and asset value loss.

Redesigned Syllabus Roadmap

CO1 - 11 HOURS

Production Planning, Plant Layout and Maintenance

PPC functions · Forecasting · Routing · Scheduling and Gantt chart · Dispatching · Production and productivity · Mass, batch, job order production · Plant location and layout · Material handling · Plant maintenance and bathtub curve

Draw: PPC flow, Gantt chart, Plant layout comparison, Bathtub curve

Calculate: Productivity

Application: Machine shop, controller-panel wiring line, tool room, material movement and maintenance planning.

CO2 - 10 HOURS

Work Study, Method Study and Work Measurement

Work study need · Method study procedure · Operation process chart · Flow process chart · Two handed process chart · Man-machine chart · Therbligs and SIMO chart · Time study · Rating factor and allowances · Standard time

Draw: Method study flow, Flow process symbols, Two-handed process chart, SIMO chart

Calculate: Normal time, Standard time

Application: Assembly work, wiring harness production, machining, packing, inspection and maintenance job standardisation.

CO3 - 12 HOURS

Quality Control and Statistical Quality Control

Quality control | Inspection types | SQC | Normal distribution | Mean median mode | Standard deviation variance | Variable and attribute data | X-bar R p c charts

Draw: Inspection flow, Normal curve, X-bar chart, R chart, p chart, c chart

Calculate: Mean, Range, Variance, Standard deviation

Application: Incoming inspection, machining dimension control, panel inspection, defect tracking and vendor quality.

CO4 - 10 HOURS

Acceptance Sampling, Estimating, Costing and Depreciation

Acceptance sampling | OC curve | Producer and consumer risk | Single double multiple plans | Estimating | Costing | Cost structure | Selling price | Depreciation methods

Draw: OC curve, Sampling flow, Cost structure ladder, Depreciation curve

Calculate: Prime cost, Factory cost, Selling price, Straight line depreciation, Diminishing balance depreciation

Application: Supplier inspection, project estimate, quotation, machine replacement and production cost control.

CO1 - 11 HOURS

Production Planning, Plant Layout and Maintenance

Machine shop, controller-panel wiring line, tool room, material movement and maintenance planning.

> PPC planning □□□□□□□ □□□□. Forecasting, routing, scheduling, dispatching, follow-up control □□□□□□□ □□□□□□□□□ production delay □□□□□□□.

PPC functions | Forecasting | Routing | Scheduling and Gantt chart | Dispatching | Production and productivity | Mass, batch, job order production | Plant location and layout | Material handling | Plant maintenance and bathtub curve

Textbook chapter explanation

PPC

PPC decides what to produce, how much, when, where and how. It links customer demand with material, machine and labour planning.

Forecasting

Forecasting estimates future demand using past data and judgement. It is the starting point for material and capacity planning.

Routing

Routing fixes the path and sequence of operations through machines and departments.

Scheduling

Scheduling fixes start and finish time. Gantt chart shows planned and actual progress.

Dispatching

Dispatching releases work orders, material issue orders, tool orders, movement orders and inspection orders.

Productivity

Productivity is output/input. It improves by better method, layout, training, maintenance and reduced rework.

Plant layout

Process, product, fixed position and combination layouts reduce movement, delay and accident.

Maintenance

Breakdown, preventive, predictive, scheduled and condition-based maintenance keep machines available. Bathtub curve shows failure pattern.

Formula / rule bank

Productivity

```
Productivity = Output / Input
```

If 120 units are made in 8 hours, productivity = 15 units/hour.

Expected questions

1. Define PPC and its functions.
2. Explain forecasting, routing, scheduling and dispatching.
3. Compare plant layouts.
4. Explain material handling equipment.
5. Explain maintenance types and bathtub curve.

Solved examples

Productivity improvement

Old output 90 pieces in 10 h, new output 120 pieces in 10 h. Old productivity = 9/h; new productivity = 12/h; improvement = 33.33%.

CO2 - 10 HOURS

Work Study, Method Study and Work Measurement

Assembly work, wiring harness production, machining, packing, inspection and maintenance job standardisation.

> Observed time □□□□□□□□ standard time □□□□□□□□. Rating factor apply □□□□□□ normal time □□□□□□□□□ allowance □□□□□□□□.

Work study need | Method study procedure | Operation process chart | Flow process chart | Two handed process chart | Man-machine chart

Textbook chapter explanation

Work study

Systematic examination of work to improve productivity. It includes method study and work measurement.

Method study

Records and critically examines existing method to develop safer and economical method.

Charts

Operation process chart, flow process chart, two-handed chart, man-machine chart, string diagram and flow diagram show work details.

Therbligs

Basic motion elements used in micro-motion study. SIMO chart records simultaneous motion.

Work measurement

Determines time required by qualified worker at defined performance level.

Time study

Break job into elements, record observed time, apply rating and add allowances.

Allowances

Personal, rest, process, special and policy allowances are added to normal time.

Formula / rule bank

Normal time

`Normal Time = Observed Time x Rating / 100`

Observed 5 min, rating 120%, normal = 6 min.

Standard time

`Standard Time = Normal Time x (1 + Allowance %)`

Normal 6 min, allowance 15%, standard = 6.9 min.

Expected questions

1. Define work study.
2. Explain method study procedure.
3. Draw chart symbols.
4. Explain two-handed chart.
5. Calculate standard time.

Solved examples

Standard time

Observed 4.8 min, rating 110%, allowance 15%. Normal = 4.8 x 1.10 = 5.28 min. Standard = 5.28 x 1.15 = 6.07 min.

CO3 - 12 HOURS

Quality Control and Statistical Quality Control

Incoming inspection, machining dimension control, panel inspection, defect tracking and vendor quality.

> Quality control defect □□□□□□□□□□ □□□□□□□ □□□□. Process stable □□□ □□□□□□ control chart □□□ □□□□□□□□□□□□□□.

Textbook chapter explanation

Quality control

Activities used to make sure product meets specification. It includes inspection, measurement, recording and correction.

Inspection

First piece, floor and centralized inspection are common types.

SQC

Statistical Quality Control uses data to separate normal variation and special-cause variation.

Normal distribution

Bell-shaped distribution where mean is centre and standard deviation shows spread.

Central tendency

Mean, median and mode describe centre of data.

Dispersion

Variance, standard deviation and range describe spread.

Control charts

X-bar chart monitors average; R chart monitors range; p chart monitors fraction defective; c chart monitors number of defects.

Formula / rule bank

Mean

```
x-bar = sum x / n
```

10,12,14 mean = 12.

Range

```
R = Maximum - Minimum
```

Max 25, min 18, range = 7.

Variance

```
Variance = sum(x-xbar)^2/n
```

Used before standard deviation.

SD

```
SD = sqrt(variance)
```

If variance = 4, SD = 2.

Expected questions

1. Define QC and objectives.
2. Compare inspection types.
3. Explain SQC.
4. Calculate mean and range.
5. Explain X-bar, R, p and c charts.

Solved examples

Mean and range

Data: 18,20,20,22,25. Mean = 105/5 = 21. Range = 25-18 = 7.

CO4 - 10 HOURS

Acceptance Sampling, Estimating, Costing and Depreciation

Supplier inspection, project estimate, quotation, machine replacement and production cost control.

> Selling price direct material + direct labour □□□□□□ □□□□. Overheads and profit □□□□ □□□□□□□□.

Acceptance sampling | OC curve | Producer and consumer risk | Single double multiple plans | Estimating | Costing | Cost structure | Selling price | Depreciation methods

Textbook chapter explanation

Acceptance sampling

A sample is inspected to accept or reject a lot. It is used when 100% inspection is costly or destructive.

OC curve

Shows probability of accepting a lot at different quality levels.

Risks

Producer risk means rejecting good lot; consumer risk means accepting bad lot.

Sampling plans

Single, double and multiple sampling plans differ by number of sampling stages.

Estimating

Predicts probable cost before production or project execution.

Costing

Determines actual/calculated cost with material, labour, expenses and overhead.

Cost structure

Prime cost, factory cost, office cost and selling price are built step-by-step.

Depreciation

Reduction in asset value due to wear, use, age or obsolescence.

Formula / rule bank

Prime cost

`Prime Cost = Direct Material + Direct Labour + Direct Expenses`

4000+2500+500=7000.

Factory cost

`Factory Cost = Prime Cost + Factory Overhead`

7000+1500=8500.

Selling price

`Selling Price = Office Cost + Selling Overhead + Profit`

9500+500+2000=12000.

Straight line

`Depreciation = (Cost - Scrap Value)/Life`

(80000-8000)/6=12000 per year.

Expected questions

1. Explain acceptance sampling.
2. Draw OC curve.
3. Compare estimating and costing.

4. Explain cost structure.
5. Calculate selling price.
6. Calculate depreciation.

Solved examples

Selling price

DM 4000, DL 2500, DE 500, FOH 1500, office OH 1000, selling OH 500, profit 2000. Prime=7000, Factory=8500, Office=9500, Selling price=12000.

Formula / Rule Bank

CO1 - Productivity

`Productivity = Output / Input`

If 120 units are made in 8 hours, productivity = 15 units/hour.

CO2 - Normal time

`Normal Time = Observed Time x Rating / 100`

Observed 5 min, rating 120%, normal = 6 min.

CO2 - Standard time

`Standard Time = Normal Time x (1 + Allowance %)`

Normal 6 min, allowance 15%, standard = 6.9 min.

CO3 - Mean

`x-bar = sum x / n`

10,12,14 mean = 12.

CO3 - Range

```
R = Maximum - Minimum
```

Max 25, min 18, range = 7.

CO3 - Variance

```
Variance = sum(x-xbar)^2/n
```

Used before standard deviation.

CO3 - SD

```
SD = sqrt(variance)
```

If variance = 4, SD = 2.

CO4 - Prime cost

```
Prime Cost = Direct Material + Direct Labour + Direct Expenses
```

4000+2500+500=7000.

CO4 - Factory cost

```
Factory Cost = Prime Cost + Factory Overhead
```

7000+1500=8500.

CO4 - Selling price

```
Selling Price = Office Cost + Selling Overhead + Profit
```

9500+500+2000=12000.

CO4 - Straight line

```
Depreciation = (Cost - Scrap Value)/Life
```

(80000-8000)/6=12000 per year.

Solved Examples / Problem Lab

CO1 - Productivity improvement

Old output 90 pieces in 10 h, new output 120 pieces in 10 h. Old productivity = 9/h; new productivity = 12/h; improvement = 33.33%.

CO2 - Standard time

Observed 4.8 min, rating 110%, allowance 15%. Normal = 4.8 x 1.10 = 5.28 min. Standard = 5.28 x 1.15 = 6.07 min.

CO3 - Mean and range

Data: 18,20,20,22,25. Mean = 105/5 = 21. Range = 25-18 = 7.

CO4 - Selling price

DM 4000, DL 2500, DE 500, FOH 1500, office OH 1000, selling OH 500, profit 2000. Prime=7000, Factory=8500, Office=9500, Selling price=12000.

Practice Section

1. PPC stands for Production Planning and Control.

Read this as an objective revision key statement.

2. Gantt chart is mainly used for scheduling.

Read this as an objective revision key statement.

3. Product layout suits mass production.

Read this as an objective revision key statement.

4. Work study includes method study and work measurement.

Read this as an objective revision key statement.

5. Standard time is normal time plus allowances.

Read this as an objective revision key statement.

6. X-bar chart monitors process average.

Read this as an objective revision key statement.

7. p chart monitors fraction defective.

Read this as an objective revision key statement.

8. Prime cost is direct material plus direct labour plus direct expenses.

Read this as an objective revision key statement.

9. Producer risk is rejecting a good lot.

Read this as an objective revision key statement.

10. Straight line depreciation gives equal annual depreciation.

Read this as an objective revision key statement.

Quick revision checklist

- CO1: Production Planning, Plant Layout and Maintenance - draw PPC flow and Gantt chart
- CO2: Work Study, Method Study and Work Measurement - draw Method study flow and Flow process symbols
- CO3: Quality Control and Statistical Quality Control - draw Inspection flow and Normal curve
- CO4: Acceptance Sampling, Estimating, Costing and Depreciation - draw OC curve and Sampling flow

Fresh Model Question Paper

Time: 3 Hours | Maximum Marks: 100 | Fresh practice paper based on supplied syllabus, not copied from any official paper.

Part A - 2 marks each

1. Define PPC.
2. What is routing?
3. State two objectives of method study.
4. Write two flow process chart symbols.
5. Define standard time.
6. What is SQC?
7. Differentiate variable and attribute.
8. What is OC curve?
9. Define prime cost.
10. What is depreciation?

Part B - 6 marks each

1. Explain functions of PPC with flow.
2. Compare mass, batch and job production.
3. Explain plant layout types.
4. Explain method study procedure.
5. Explain method study charts.
6. Calculate standard time from observed time, rating and allowance.
7. Explain X-bar, R, p and c charts.

Part C - 10 marks each

1. A job has observed time 8 min, rating 125%, allowance 20%. Calculate standard time.
2. For 12,15,14,16,13 calculate mean and range.
3. Explain acceptance sampling, OC curve and risks.
4. Calculate selling price from direct costs, overheads and profit.
5. Calculate depreciation by straight line method.

Complete Model Answers

Standard time

Normal time = 8 x 125/100 = 10 min. Standard time = 10 x 1.20 = 12 min.

Mean and range

Mean = (12+15+14+16+13)/5 = 14. Range = 16-12 = 4.

Selling price structure

Prime cost = direct material + direct labour + direct expenses. Factory cost = prime cost + factory overhead. Office cost = factory cost + office overhead. Selling price = office cost + selling overhead + profit.

Depreciation

Straight line depreciation = (Cost - Scrap value) / Life.

OC curve

OC curve shows probability of accepting a lot at different quality levels. Producer risk is rejecting a good lot. Consumer risk is accepting a bad lot.

References

- M. Mahajan, Statistical Quality Control, Dhanpat Rai Publishing Co Pvt Ltd.
- O.P. Khanna, Industrial Engineering and Management, Dhanpat Rai Publications.
- R. Keith Mobley, Maintenance Fundamentals, 2nd Edition, Elsevier.
- NPTEL and SWAYAM online resources listed in the syllabus.